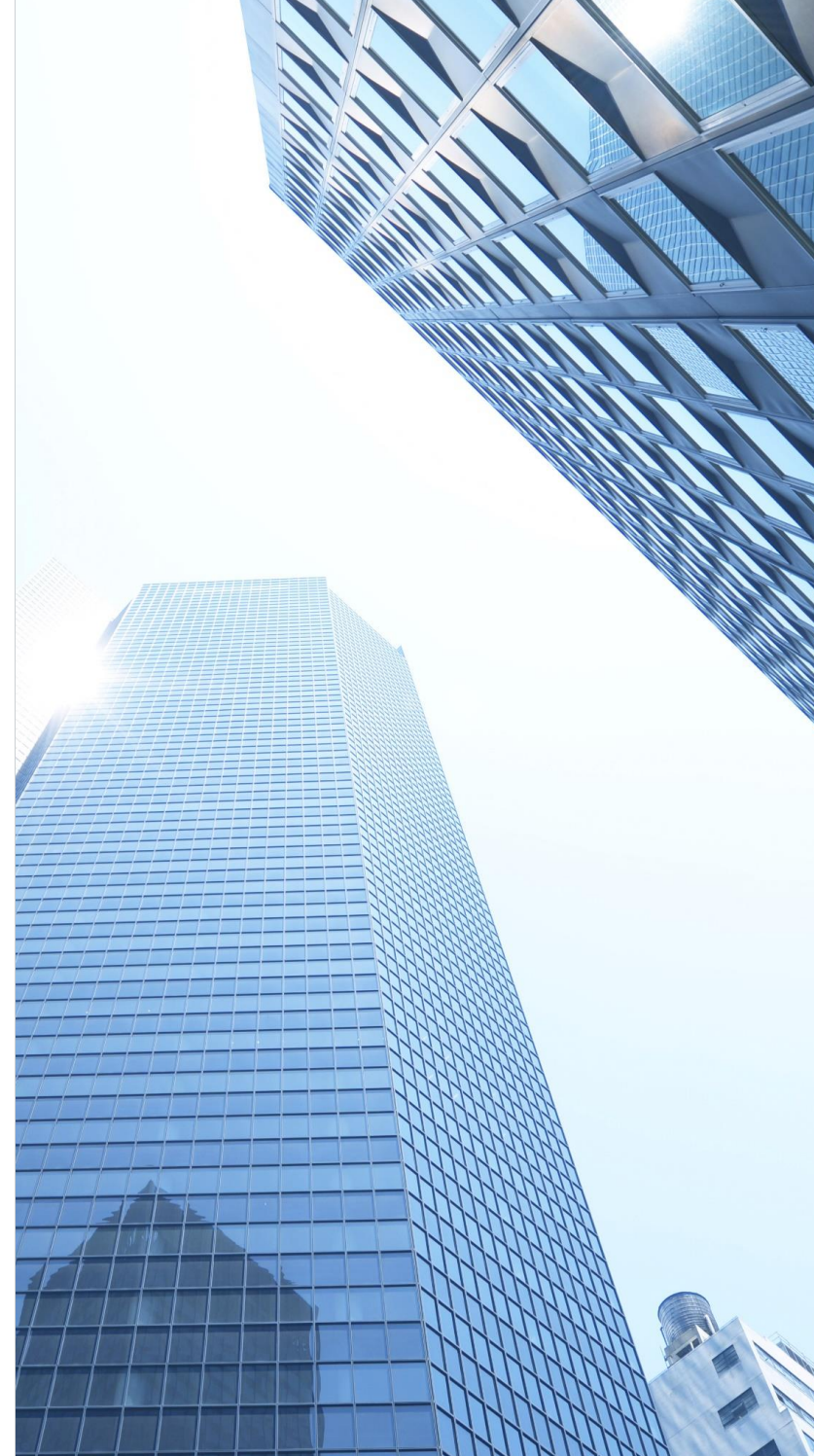




Polo to the M5: VW, BMW switch utilising cointegration

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2026 Global Markets Intern

Private and confidential





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1. Trade Idea
2. What is stationarity and cointegration?
3. How can we use cointegration for trading?
4. My research process
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6. Catalysts and risks
7. Execution strategy

Client Buy

Bond	Issuer	Benchmark-spread ask	Z-spread ask	YTM ask	Size
BMW 3.375 08/34	BMW International Investment B.V.	87	83	3.95	€10mn

Client Sell

Bond	Issuer	Benchmark-spread bid	Z-spread bid	YTM bid	Size
VW 3.5 06/31	Volkswagen Bank GmbH	97	77	3.83	€10mn

Metrics

- +12 YTM, +5 Z, -3 G
- BBB+/A1 to A/A2
- SP to Senior unsecured
- Both €700mn+ outstanding
- Both issued past 2024
 - +2 duration

My argument

Statistically and fundamentally, the difference in spread between the bonds is an anomaly

This anomaly has primarily occurred due to bond technicals

Client profits from mean reversion occurring

Target clientele

Hedge Funds

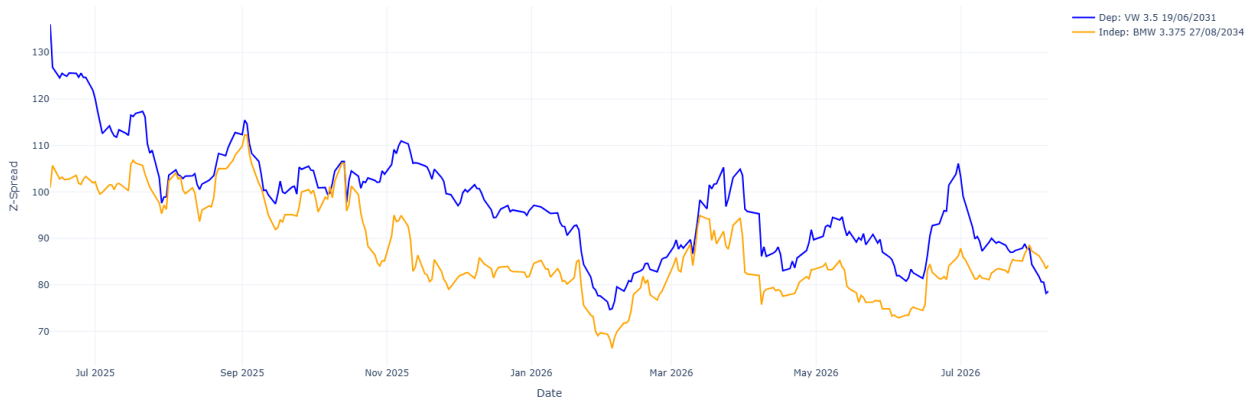
What is stationarity and cointegration?

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

Z-Spreads: Dep VW 3.5 19/06/2031 vs Indep BMW 3.375 27/08/2034

Stationarity: A time series, A_t , has a constant mean and standard deviation over time

Cointegration: There exists constants α and β such that $\varepsilon_t = A_t - (\alpha + \beta B_t)$ is stationary



VW 3.5 and BMW 3.375 z-spreads over time

OLS Z-Spread Regression: Dep VW 3.5 19/06/2031 vs Indep BMW 3.375 27/08/2034

Variable	Interpretation
A_t, B_t	Bond Z-spread
α	Average spread difference
β	Hedge ratio
ε_t	Residual
$\alpha + \beta B_t$	Predicted A_t
$A_t - \beta B_t$	Beta - Adjusted spread difference

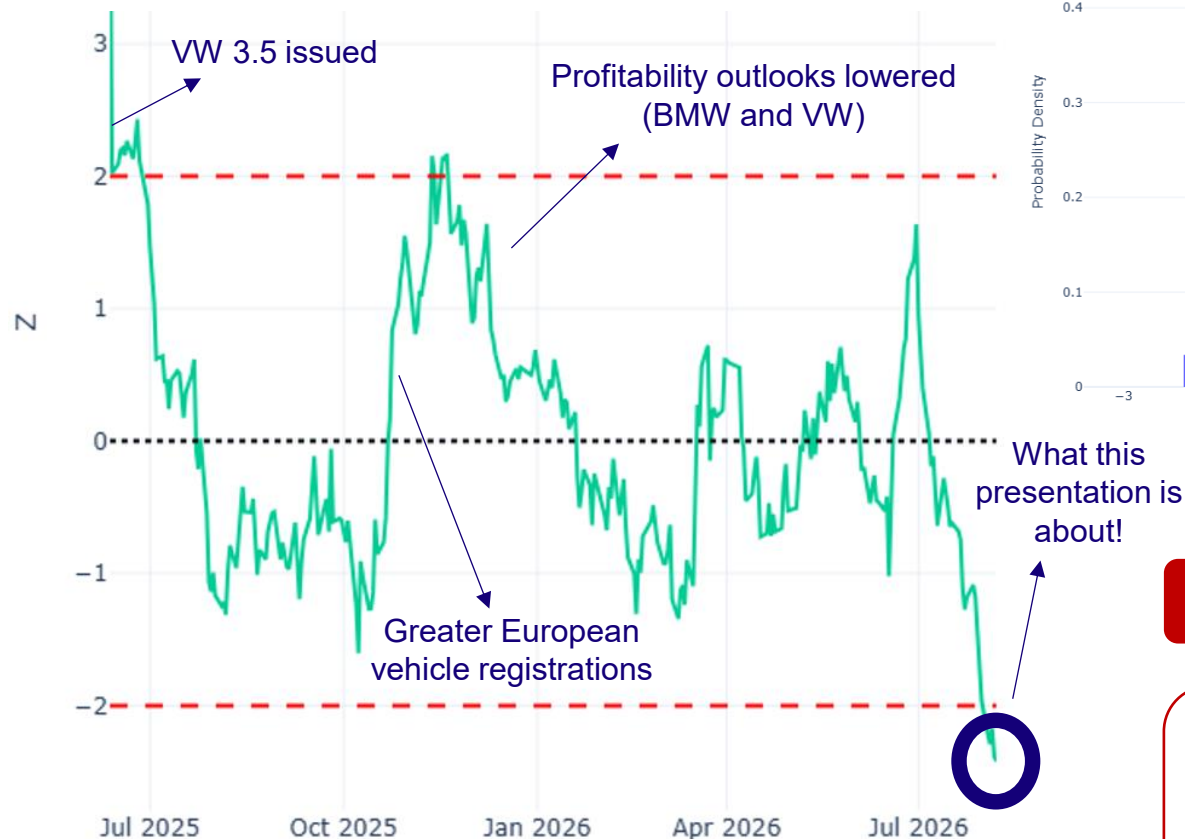


VW 3.5 and BMW 3.375 linear regression

How can we use cointegration to develop trading signals?

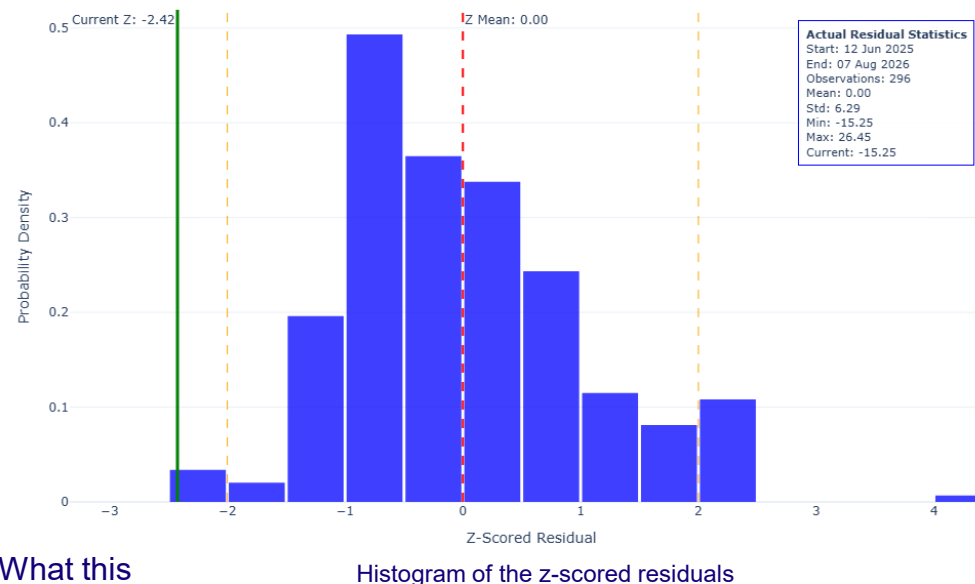
Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

A stationary time series fluctuates about a mean, so detecting large residuals is a possible signal for a mean reverting trade to be put on



Plot of the z-scored residuals, $A_t = \text{VW Z-spread}$,
 $B_t = \text{BMW Z-spread}$, $\alpha = 14.59$, $\beta = 0.94$, $R^2 = 0.7$
Residual: $\varepsilon_t = A_t - (\alpha + \beta B_t)$, Spread: $A_t - B_t \approx \alpha + \varepsilon_t$

Z-Scored Residual Distribution: Dep VW 3.5 19/06/2031 vs Indep BMW 3.375 27/08/2034

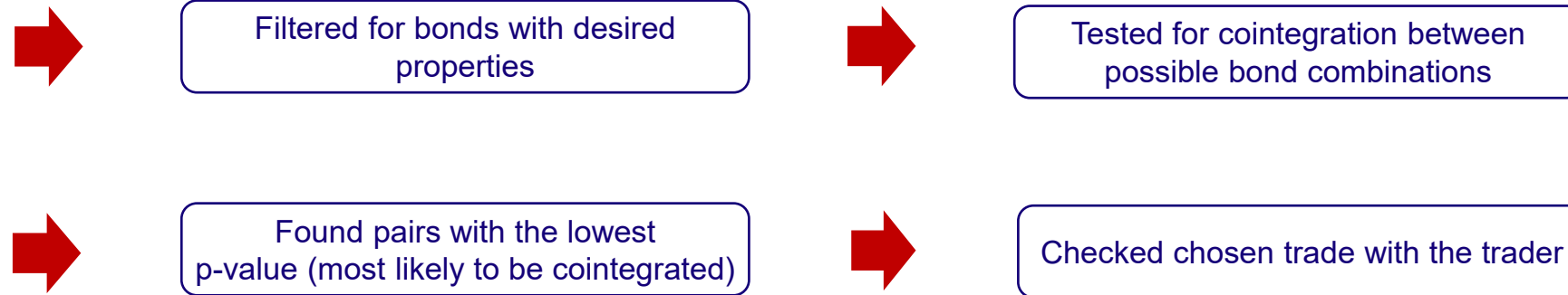


Cointegration Model Risk

- Market regimes can change
- False positives in the ADF hypothesis test
- α and β can vary slightly over time with new data points

My research process

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31



Top pairs

Current	Dep: BMW 3.375 02/02/2034 vs Indep: BMW 3.5 17/11/2032	Z: 2.68 Alpha: 7.66 Beta: 1.05 r_squared: 0.94 Current Spread: 16.44 Current - Mean Spread: 8.79
Current	Dep: BMW 3.375 02/02/2034 vs Indep: BMW 3.625 22/05/2035	Z: 2.65 Alpha: 11.13 Beta: 0.90 r_squared: 0.93 Current Spread: 21.10 Current - Mean Spread: 9.97
Current	Dep: BMW 3.75 20/11/2034 vs Indep: RENAUL 4.125 04/04/2031	Z: 2.51 Alpha: 20.89 Beta: 0.66 r_squared: 0.83 Current Spread: 35.30 Current - Mean Spread: 14.42
Current	Dep: VW 3.5 19/06/2031 vs Indep: BMW 3.375 27/08/2034	Z: -2.42 Alpha: 14.59 Beta: 0.94 r_squared: 0.70 Current Spread: -0.66 Current - Mean Spread: -15.25
Current	Dep: VW 3.5 19/06/2031 vs Indep: RENAUL 3.875 30/09/2030	Z: -2.32 Alpha: 20.63 Beta: 0.78 r_squared: 0.85 Current Spread: 10.26 Current - Mean Spread: -10.37
Current	Dep: TOYOTA 3.125 11/07/2029 vs Indep: VW 4.0 11/04/2031	Z: -2.20 Alpha: 5.24 Beta: 0.49 r_squared: 0.87 Current Spread: -4.91 Current - Mean Spread: -10.15
Current	Dep: BMW 4.125 04/10/2033 vs Indep: BMW 3.375 02/02/2034	Z: -2.18 Alpha: -0.74 Beta: 0.92 r_squared: 0.92 Current Spread: -9.28 Current - Mean Spread: -8.54
Current	Dep: BMW 3.125 27/08/2030 vs Indep: MBGGR 3.25 10/01/2032	Z: -2.16 Alpha: -9.80 Beta: 1.01 r_squared: 0.95 Current Spread: -17.45 Current - Mean Spread: -7.65
Current	Dep: TOYOTA 3.125 11/07/2029 vs Indep: VW 4.625 25/03/2029	Z: -2.16 Alpha: 17.47 Beta: 0.47 r_squared: 0.88 Current Spread: 7.84 Current - Mean Spread: -9.64
Current	Dep: VW 3.5 19/06/2031 vs Indep: BMW 3.5 17/11/2032	Z: -2.14 Alpha: 6.70 Beta: 1.16 r_squared: 0.83 Current Spread: -3.39 Current - Mean Spread: -10.10

The chosen trade idea



Thesis 1: BMW International Investment is the 'lower risk' entity

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

		
Business complexity	Lower An internal treasury and funding department	Higher A bank that handles retail and dealer financing, leasing and deposit businesses
Counterparties	BMW Group Companies Raises money and lends to BMW group companies	Retail, dealers and VW Group Companies Lends to retail, dealers and VW Group entities
Credit rating	A-/A2 'The ratings are supported by BMW's strong balance sheet'	BBB+/A1 'Residual value risks from the large and growing lease portfolio'
Credit provision (2025)	€1mn Loans and receivables total to €10.8bn	€714mn Loans and receivables total to €96.4bn
Debt Guarantor	BMW AG Debt issued is backed by BMW AG	None Debt issued is not backed by another VW entity

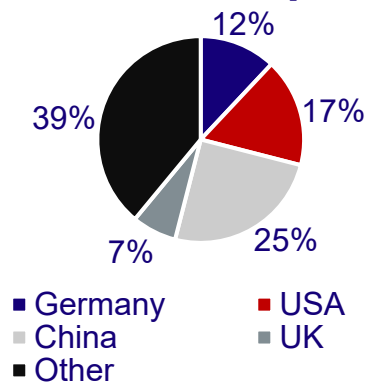
Thesis 2: VW Bank is more susceptible to the macro environment

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

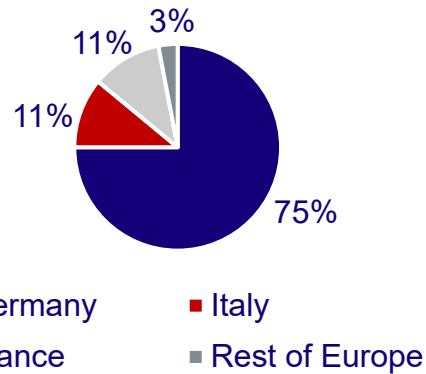
Revenue by region (2025)



BMW Group*

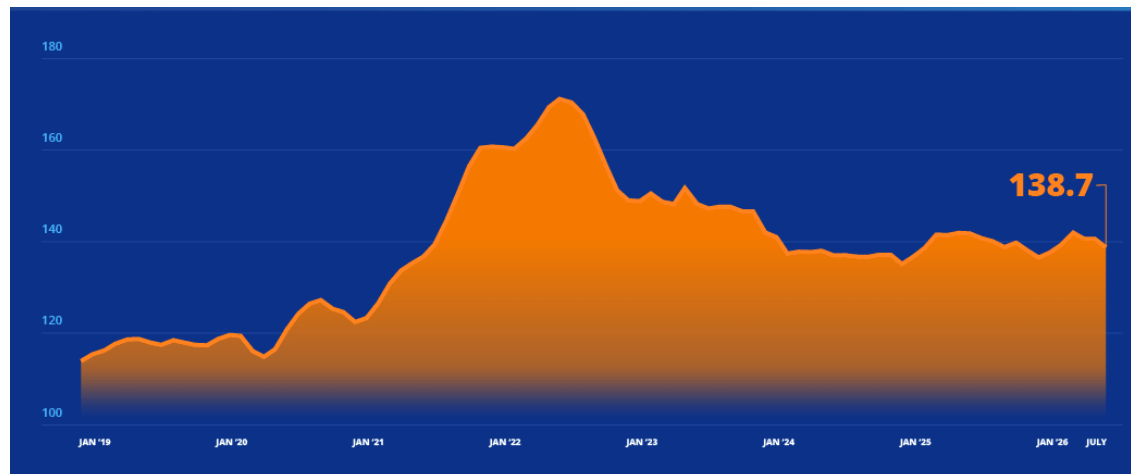


VW Bank

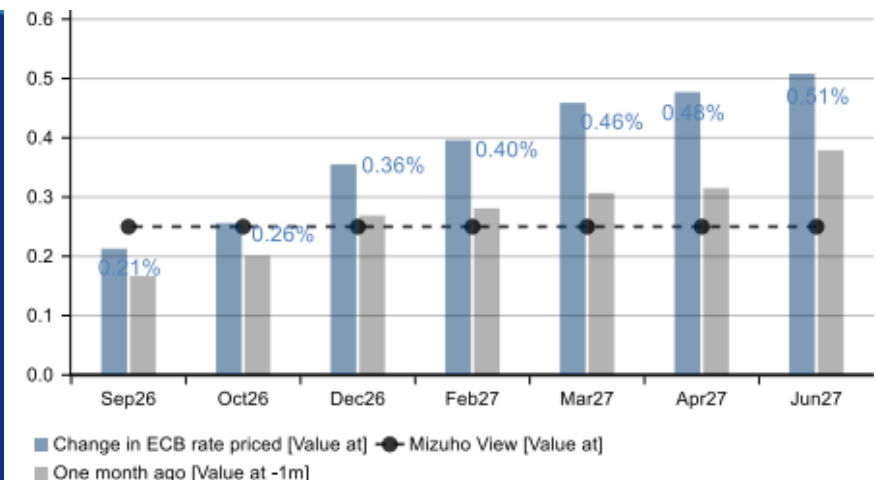


*Is used as a proxy for the geographical exposure of BMW funding companies

- Germany's unemployment rate is rising, currently at 6.4%
- Euro area energy inflation is up 10% year on year
- Potential ECB rate hikes with market priced for an October hike
- Chinese car brands accounted for 14% of European EV sales in 2026 so far in despite of tariffs



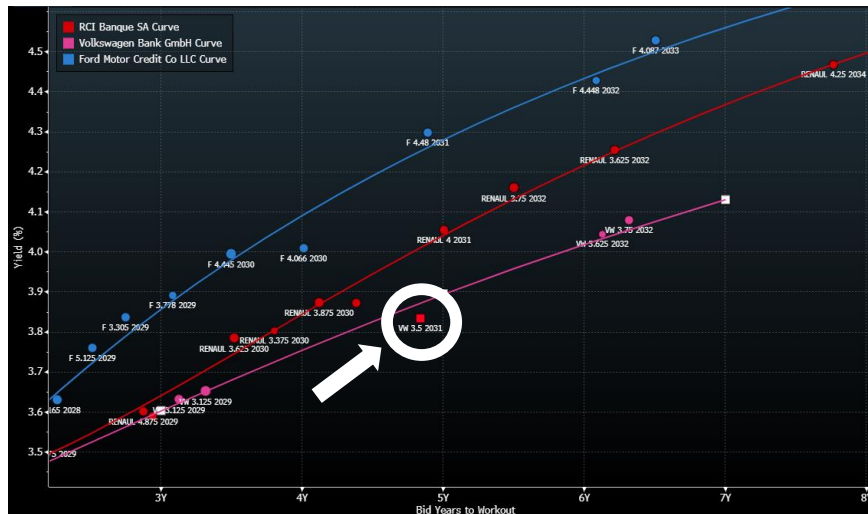
AUTO1 Group Price Index – European Used Car Price report



Market implied rate changes and Mizuho ECB rate views

Thesis 3: VW 3.5 31 is an outperformer

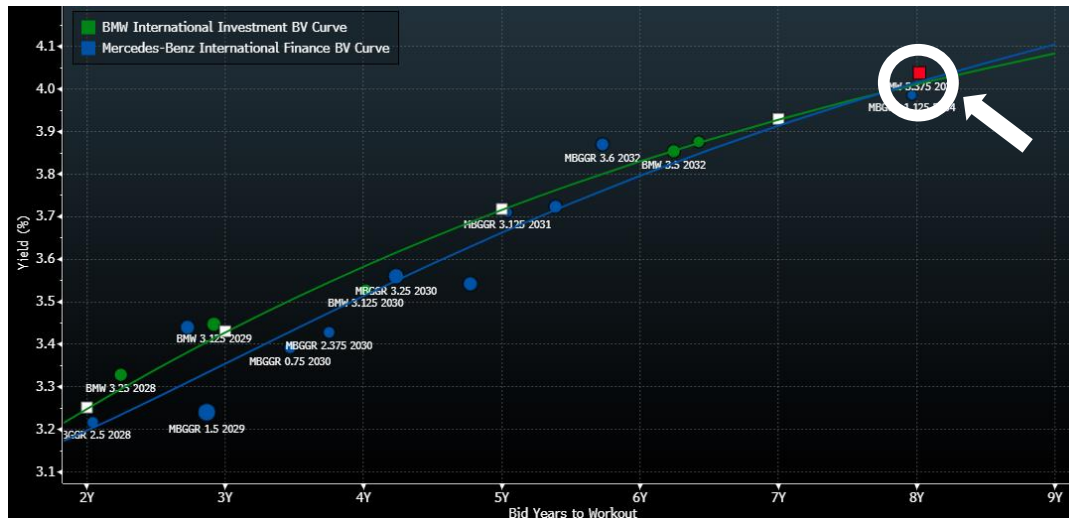
Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31



VW Bank (pink), RCI Banque (red) and Ford Motor Credit (blue) yield curves



VW 3.5 31 yield vs Bloomberg Euro-Aggregate Automotive Index yield

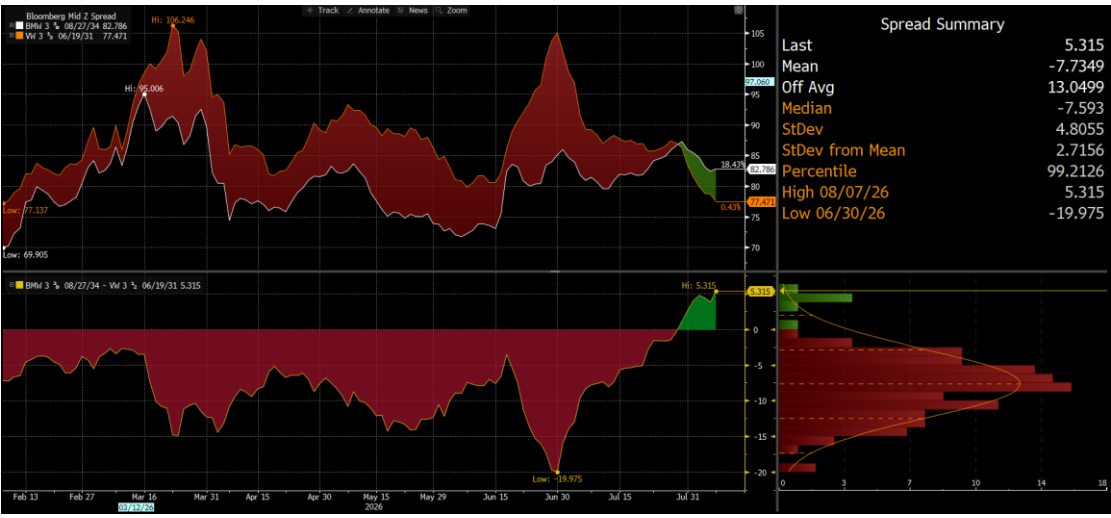


BMW International Investment (Green) and Mercedes Benz-International Finance (Blue) yield curves

- The VW bond YTM has decreased 19bp from its highs this month
- BMW bond is underperforming relative to its curve

Thesis 3: VW 3.5 31 is an outperformer

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

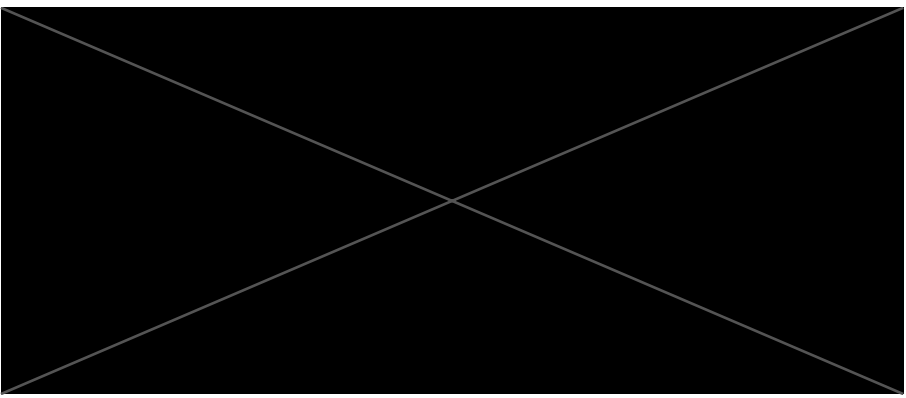


VW 3.5 31 vs BMW 3.375 34 mid Z-spread

VW 3.5 31 vs VW Group of America Finance issues

	Bond	Z
USD Bonds	VW 3.5 08/31	104*
	VW 6.45 11/30	110
	VW 5.35 03/30	112
	VW 3.75 05/30	115
	VW 4.85 09/30	114
	VW 5.65 03/32	136

*Utilising XCCY Z-spread



- Real money buying has pushed the VW price up and the street short
- Street trying to buy back shorts, led to further increases in price



Z-scored residuals between the VW 3.5 31 and BMW 3.375 34

Catalysts

- **Real money taking profit on their positions**
 - ECB rate hike
 - Potential VW issuance

Risks

- **Negative BMW Capital Markets day (September 29th and 30th)**
 - Real money holding on to bonds

Execution strategy

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

Trade details:

Timeframe: 2 Months, Size: 10mn x 10mn, BMW 3.375 DV01: 659 on €1mn, VW 3.5 DV01: 430 on €1mn

Base case

Long leg:

Target spread tightening: 3bps

P&L = + €19770

Short leg:

Target spread widening: 7bps

P&L = + €30100

Total expected profit
€50k

Bull case

Long leg:

Target spread tightening: 5bps

P&L = + €32950

Short leg:

Target spread widening: 10bps

P&L = + €43000

Total expected profit
€77k

Bear case

Long leg:

Target spread tightening: 1bps

P&L = + €6590

Short leg:

Target spread widening: 4bps

P&L = + €17200

Total expected profit
€24k

Stop loss:

Long leg:

Spread widening: 2bps

P&L = - €13180

Short leg:

Spread tightening: 2bps

P&L = - €8600

Total expected loss
- €22k

Appendix: Repo Calculations

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

Long leg

P&L = €19770 (ignoring financing)

Net yield = $3.93\% - 3.05\% = 0.83\%$

(Mizuho) Funding cost = -2.46%

Reverse Repo = 2.19%

Net Carry = 0.56%

Carry = $10\text{mn} \times 0.17 \times 0.56\% = \text{€}9333$

P&L = + €29103 (using Mizuho financing rates)

**Base case P&L = + €41k
(using Mizuho financing rates)**

Short leg

P&L = €30100 (ignoring financing)

Net yield = $-3.78\% + 2.86\% = -0.92\%$

Repo = -2.19%

Reverse Repo = 2%

Net Carry = -1.11%

Cost = $10\text{mn} \times 0.17 \times -1.11\% = -\text{€}18500$

P&L = + €11600 (using Mizuho financing rates)

Appendix: Backtest results on cointegrated beta-adjusted bond pairs

Buy: BMW 3.375 08/34
Sell: VW 3.5 06/31

Auto Credit Bond Pairs Portfolio

